

Max Carroll

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EDUCATION

University of Cambridge Sidney Sussex College <i>Master of Engineering in Computer Science</i> Honours: Distinction Award: 1st Place Dissertation (93%)	Cambridge, UK Oct. 2025 – Jul. 2026 Rank: 2nd (87.4%)
University of Cambridge Sidney Sussex College <i>Bachelor of Arts in Computer Science</i> Honours: First Class with Distinction Awards: 2nd Place Dissertation (92.5%), Frank George Prize, Philip Haswell Scholarship	Cambridge, UK Oct. 2022 – Jul. 2025 Rank: 6/134 (84%)
Lawrence Sheriff School 4 A*s in Mathematics, Further Mathematics, Physics, and Music	2015 – 2022

SKILLS

<i>Secondary knowledge of 15+ additional programming languages.</i>	
Languages:	OCaml, Haskell, Agda, C#, F#, Python, Java, C, SQL
Technical:	Programming Language/DSL Design, Compilers & Interpreters, AI & ML, Proof Assistants, Type Systems & Theory, HCI, Data Structures, Discrete Mathematics, Category Theory, Abstract Algebra
Tools:	Git, Unix Tools, GitHub Actions, Nix
Personal:	Self-Learning, Focus, Communication Skills, Leadership & Teamwork, Research Skills, Presentation Skills

MAJOR PROJECTS, RESEARCH & EXPERIENCE

Research Internship <i>Devised Type Slicing through independent research (see below)</i>	University of Cambridge, UK Jun. 2025 – Oct. 2025
Research Internship <i>Conducting Type Slicing research and implementation with a global research team</i>	University of Cambridge, UK Jun. 2026 – Present
Type Slicing OCaml, Agda Type Systems, Debugging Tools, HCI, Lattices, Proofs <i>Soft skills: Self-Learning, Research Skills, Presentation Skills</i> - Conducted research into type debugging, static analysis, formal methods, and HCI. - Devised a lattice-based theory of Type Slicing with machine-checked correctness proofs. - Presented at the world-leading ICFP/SPLASH conference; further publications are under review. - Identified applications in live programming and integrated LLM assistants. - Implemented Type Slicing in <i>Hazel</i> with an international research team.	
AI Kayak Hull Optimisation Python AI (Gaussian Processes), Parameterisation, Graphics <i>Soft skills: Leadership, Teamwork, Presentation Skills</i> - Applied Gaussian-process optimisation to improve kayak-hull stability. - Supported practical constraints and target stability characteristics. - Co-designed the system with kayakers and a small technical team.	Team
Blind Music Composition Assistant Java DSL Design, AI (LLMs), HCI, Parsing <i>Soft skills: Leadership, Teamwork, Communication Skills, Presentation Skills</i> - Built an accessible assistant using LLMs to map speech to a music-editing DSL. - Led client meetings, requirements gathering, and team coordination. - Designed the DSL using my experience as a composer.	Team

INTERESTS AND OTHER ACHIEVEMENTS

UKMT Senior Mathematical Challenge Gold and progression to the Senior Kangaroo	2021
Music: Diploma-standard pianist Flamenco guitarist Cellist Classical-music composer	
Sports: Football and tennis team captain	

REFEREE

Professor Anil Madhavapeddy	<i>References available upon request</i>
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